

FIELDLINK: TACTICAL MESH OPERATIONS

OFFLINE-FIRST DECENTRALIZED ASSET, MUSTER & SITREP COMMAND PLATFORM

Technical Architecture, Protocol Specifications & Technology Stack Deep Dive · Version 2.0

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

During severe disaster response, remote tactical deployments, or subterranean missions, traditional cellular and satellite backhubs fail or experience severe electromagnetic jamming. FIELDLINK (Offline-Mesh-Asset-Sync) is an autonomous, mission-critical Progressive Web Application (PWA) designed to provide 100% operational uptime without internet access. Field operators can coordinate assets, muster personnel, and share real-time intelligence through a decentralized mesh network. The platform features a 3D spatial network map, GIS map integration, and a command interface for asset tracking and communication. It supports offline-first operation, ensuring critical data is preserved and synchronized when connectivity is restored. The system is built using modern web technologies, including React, TypeScript, and Vite, and is designed for deployment on various hardware platforms, including ruggedized mobile devices and dedicated mesh routers.

2. COMMAND & 3D VISUALIZATION

2.1 HYPERMESH OPERATIONS

Frontend UI & React Core: React 19.2, TypeScript 6.0, Vite 8.3 (Fast HMR & ESM chunking)

Spatial Visualization: Three.js 0.186, OrbitControls, 3D Raycasting, Canvas Billboard Sprites

Design System & Audio: Tailwind CSS 3.4 (Light Tactical Theme), Lucide Icons, Web Audio API

Local Persistence Layer: Browser IndexedDB, Dexie.js, LocalStorage snapshots, Service Worker PWA

P2P Mesh Transports: WebRTC (RTCPeerConnection), Web Bluetooth API (BLE), Local LAN, Cloud Relay

Conflict Resolution (CRDT): State-based Last-Write-Wins (LWW), Hybrid Logical Clocks (HLC), Vector Clocks

Security & Integrity: Web Crypto API (SubtleCrypto), AES-GCM 256-bit encryption, SHA-256 Block Hashes

Verification & Testing: Vitest 5.0, Oxlint, In-App Interactive Automated CRDT Verification Test Bench

DECENTRALIZED TRANSPORT & CRDT ENGINE

4. MULTI-TRANSPORT MESH ARCHITECTURE

FIELDLINK does not rely on a single physical medium. The TransportManager coordinator abstracts heterogeneous network layers and dynamically selects the highest bandwidth, lowest latency transport available:

1. WebRTC DataChannels (P2P Direct)

Zero-relay direct peer browser streams with sub-20ms latency. Transfers binary CRDT operation batches directly between devices on shared subnets or via lightweight signaling.

2. Web Bluetooth Low Energy (BLE)

Short-range hardware beacons for air-gapped field squads where radio silence or zero-infrastructure environment is required. Transmits compact binary deltas (MTU 512 bytes).

3. Local Network LAN / WebSocket

High-throughput tactical base-station synchronization across tactical local area networks, WiFi access points, or rugged field switches.

4. Cloud Relay Fallback

Secure WebSocket relay hub used when backhaul connectivity is intermittently restored to bridge isolated field sectors back to Headquarters.

5. CONFLICT-FREE REPLICATED DATA TYPE (CRDT) ENGINE

FIELDLINK guarantees Strong Eventual Consistency (SEC) across all nodes without requiring centralized locking or coordinator elections:

- **Hybrid Logical Clocks (HLC):** Combines physical monotonic timestamps with logical Lamport counters to generate causally ordered, globally unique mutation identifiers (Format: <timestamp>-<counter>-<deviceId>).
- **Last-Write-Wins (LWW) Register:** Deterministically resolves field-level conflicts across divergent branches. In the rare event of identical HLC timestamps, device ID byte-ordering breaks ties uniformly.
- **Merkle Tree Delta Synchronization:** Exchanges cryptographic tree roots between peers to identify missing mutations in $O(\log N)$ comparisons, drastically conserving radio battery and bandwidth.
- **Tamper-Evident SHA-256 Audit Trail:** Every state mutation and handshake event is chained into an immutable SHA-256 ledger, preventing spoofing and providing auditability.

6. INTERACTIVE 3D TACTICAL MESH VISUALIZER

The spatial network visualizer is powered by Three.js with hardware-accelerated WebGL rendering. Key features include OrbitControls (orbit, pan, zoom with safe clipping), real-time Raycasting for mouse hover and click selection, floating dynamic sprite billboard labels, animated packet propagation along multi-hop routes, and a rich telemetry inspector panel detailing RSSI, latency, battery levels, and trusted status.

OPERATOR RUNBOOK: HOW TO USE THE PROTOTYPE

Follow this step-by-step operational guide to test, demonstrate, and leverage the FIELDLINK prototype to its fullest capability:

STEP 1: COMMAND DASHBOARD & 3D MESH EXPLORATION

- View the top summary bar answering all 8 operational readiness questions at a glance.
- Use the mouse inside the 3D canvas: Left-drag to rotate the mesh, Right-drag to pan, Scroll wheel to zoom.
- Hover over any node (e.g. A-17, B-04, C-12, R-01, M-08) to see glowing link highlights and status tooltips.
- Click a node to open the rich Telemetry Inspector displaying RSSI, battery, latency, and transport type.
- Click "Ping Mesh" to dispatch a multi-hop test packet sweep across all active links with latency feedback.

STEP 2: DEMO NETWORK FAILURE SIMULATION (ZERO-BANDWIDTH TEST)

- Toggle DEMO MODE from the top header or settings.
- In the Network Failure Simulator bench on the dashboard, toggle "Internet / Cloud" to OFFLINE.
- Disconnect Peer B-04 or Node M-08 to simulate radio isolation.
- Notice how FIELDLINK stays 100% responsive, queues operations locally, and displays the offline badge.

STEP 3: ASSET & EQUIPMENT ACCOUNTABILITY

- Navigate to "Assets & Equipment" to search, filter by category/condition, and check out items.
- Click "+ New Asset" to create gear or use the QR Scanner Simulator to simulate scanning barcodes.
- Change asset status from "Available" to "Deployed" to trigger automatic CRDT mutation generation.

STEP 4: PERSONNEL MUSTER & ROLL CALL SESSIONS

- Navigate to "Personnel & Muster" to review tactical fireteam accountability.
- Click "Start Roll Call Session" to conduct live attendance (Present, Absent, Missing, Injured).
- Submit the session to write an immutable roll-call record and update the composite readiness score.

STEP 5: TACTICAL CHECKLISTS & SITREP INCIDENT LOGGING

- Navigate to "Checklists" to execute pre-flight or trauma triage verification with live progress rings.
- Navigate to "Incidents / SITREP" or click "Report SITREP" on Dashboard to broadcast urgent casualty or perimeter alerts with coordinates and severity ratings.

STEP 6: SYNC CENTER, VERIFICATION SUITE & VAULT BACKUP

- Navigate to "Sync Center" to inspect the 8-stage convergence pipeline, CRDT queues, and peer links.
- Click "Verify CRDT" in the top header to run the built-in automated test suite verifying deterministic convergence.
- In "Settings", export full encrypted JSON vault snapshots for cold backup or air-gapped migration.